

Gaussian4XWAM: Gaussian Field Enhancement for World-Action Models

Abstract—World Action Models (WAMs) and video-action models offer a promising alternative to reactive Vision-Language-Action policies by coupling action generation with future visual dynamics. However, most existing WAMs still represent the world in RGB frames or two-dimensional video latent spaces, which can be visually coherent yet geometrically under-constrained for manipulation. We propose Gaussian4XWAM, a training-only Gaussian field enhancement framework for WAMs. Gaussian4XWAM constructs a 3D Gaussian teacher from multi-view observations and renders dense feature, depth, alpha/mask, and validity targets to supervise action-relevant latent tokens. The Gaussian teacher is used only during training and is fully removed at inference time, preserving the original policy interface and adding no test-time 3D reconstruction or rendering overhead. Experiments on LIBERO and LIBERO-Plus show that Gaussian4XWAM improves FastWAM from 96.6% to 97.6% on LIBERO and from 52.05% to 70.99% overall on LIBERO-Plus. On Cosmos Policy, Gaussian4XWAM improves the LIBERO-Plus overall success rate from 71.52% to 77.30%. Ablations further show that dense 3D feature, depth, alpha supervision, and the supervised layer choice all affect robustness, supporting the value of training-time 3D geometric grounding for world-action learning.

I. INTRODUCTION

Recent World Action Models (WAMs) and video-action models have introduced a new modeling paradigm for robotic control [27], [25], [26]. Unlike conventional Vision-Language-Action (VLA) models that directly predict actions from current observations [1], [2], these methods aim to unify future visual dynamics and action generation by leveraging the temporal dynamics and implicit physical priors embedded in video generative models. As illustrated in Fig. 1, this shift moves policy learning from purely reactive action prediction toward action-conditioned world evolution modeling.

However, existing WAMs typically represent the world primarily in RGB image space or two-dimensional video latent space. Such 2D video representations can capture appearance changes, motion continuity, and certain implicit dynamic regularities, but the physical world in which robots operate is inherently three-dimensional. Successful manipulation often depends on metric depth, 3D object layout, contact relations under occlusion, and the spatial reachability between the robot and surrounding objects. Relying solely on image reconstruction or 2D video prediction may therefore produce latent representations that are visually coherent but geometrically under-constrained, limiting the precision and stability of action generation [19], [20], [21], [22], [17], [18], [23]. A direct solution is to extend WAMs into explicit 3D or 4D world models that predict future depth, point clouds, 3D scene representations, or multi-view geometric

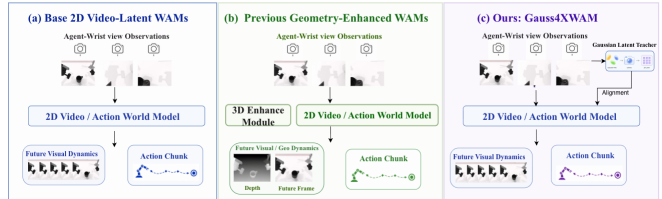


Fig. 1. Comparison of robotic control paradigms. Conventional VLA models directly map observations to actions, while WAMs and video-action models incorporate future visual dynamics into action generation. Gaussian4XWAM enhances WAMs with training-time 3D Gaussian supervision, without changing the inference pipeline.

structures [24], [19]. While such designs can improve spatial grounding, they also introduce substantial costs. High-quality 3D robot interaction data are far more scarce than internet-scale images and videos, making it difficult to acquire robust 3D physical understanding through data scaling alone. Moreover, explicit 3D/4D modeling often requires additional geometric supervision, 3D encoders and decoders, rendering modules, or future 3D rollouts, which increase training complexity and may also introduce extra memory usage and latency during inference. These costs can weaken the efficiency advantage of WAMs in real-time closed-loop control. This raises a practical question: can we use explicit 3D geometric supervision only during training to shape the latent representations of WAMs, while preserving the original inference interface and computational efficiency at test time?

To this end, we propose **Gaussian4XWAM**, a **Gaussian Field Enhancement for World-Action Models**. Rather than building a full 3D world model or requiring WAMs to generate future 3D scenes during training or inference, Gaussian4XWAM uses a 3D Gaussian field as a training-time geometric teacher to enhance action-relevant latent tokens in existing WAMs.

Different from directly using 2D features from visual foundation models such as VGGT [12], CLIP [13], DINO [14], or SAM [15], a Gaussian field provides an explicit 3D-structured intermediate representation [30], [28], [29]. Two-dimensional feature maps are often tied to specific camera viewpoints and lack cross-view geometric consistency, occlusion reasoning, and visibility constraints. In contrast, a Gaussian field aggregates multi-view semantic, geometric, and appearance information into 3D Gaussian primitives [16], which can then be rendered or projected back to the 2D token grid of the original video-action backbone. This preserves the semantic strength of 2D foundation models while injecting depth, spatial layout, alpha/mask, and visibility constraints

into the 2D video latent space.

Concretely, we construct a Gaussian field from current multi-view observations and project it into dense supervision signals aligned with the WAM backbone. These signals guide the action-relevant latent representation toward stronger spatial consistency and geometry-aware action understanding. During inference, the Gaussian teacher is completely removed, and the model follows the original WAM inference pipeline without any additional 3D reconstruction, rendering, or rollout overhead.

Our contributions are summarized as follows:

- We propose **Gaussian4XWAM**, a Gaussian teacher enhanced world-action learning framework that can be plugged into existing WAMs and video-action models to inject 3D geometric supervision into action-relevant latent representations.
- We design a **training-only 3D enhancement** mechanism that uses Gaussian fields to provide dense feature, depth, alpha/mask, and visibility-aware supervision, while completely removing the Gaussian teacher at inference time. This preserves the original WAM inference workflow and introduces no additional test-time cost.
- We validate Gaussian4XWAM on two representative WAM architectures: Cosmos Policy, a unified transformer-based model, and FastWAM, a dual-expert model. Experiments on LIBERO and LIBERO-Plus demonstrate consistent improvements, showing that Gaussian4XWAM enhances 3D spatial grounding, robustness, and action generation.

II. METHOD

A. Overview

We propose **Gaussian4XWAM**, an architecture-agnostic 3D Gaussian teacher framework for world action models (WAMs). Given robot visual observations, language instructions, and proprioceptive states, existing WAMs usually learn future states and actions in 2D video latents or world tokens. Such 2D representations are effective, but they may lack explicit and stable 3D spatial constraints, which can hurt generalization under viewpoint changes, occlusion, and precise manipulation.

As shown in Fig. 2, Gaussian4XWAM addresses this by constructing an offline 3D Gaussian teacher from robot observations. The teacher is rendered into dense training targets and used to supervise internal world representations of a WAM. The teacher is used only during training; at inference time, all Gaussian teacher modules are removed, and the original WAM forward path is unchanged. We apply the same teacher framework to two representative architectures: a FastWAM-style dual-expert MoT model [8] and a Cosmos-Policy-style unified Transformer model [9].

B. Gaussian Teacher Construction

For each training sample, we construct a 3D-aware dense teacher from synchronized multi-view images $\{I^v\}_{v=1}^V$. The current teacher construction uses visual observations only;

language instructions remain part of the policy input but are not fed into the Gaussian teacher.

First, a frozen geometry foundation model, VGGT-Omega [12], predicts depth, confidence, and camera parameters for each view:

$$\{D^v, C^v, K^v, E^v\}_{v=1}^V = \text{VGGT}(\{I^v\}_{v=1}^V). \quad (1)$$

Valid depth pixels are back-projected into 3D points using the predicted intrinsics and extrinsics. Low-confidence or invalid pixels are filtered out to form a geometry valid mask.

Second, a frozen CLIP vision encoder [13] extracts dense patch-level visual features. We use only the CLIP image branch, not the text branch. The patch tokens are reshaped and resized to the same grid as the geometry map, producing a dense feature target F_{vis}^v . In this role, CLIP is used similarly to a DINO-style dense visual feature teacher [14].

We then initialize a 3D Gaussian field from the valid 3D points. Each Gaussian primitive contains geometry, opacity, color, and a feature vector:

$$g_i = \{x_i, s_i, \alpha_i, c_i, z_i\}, \quad (2)$$

where z_i is initialized from the dense CLIP feature at the corresponding pixel. The Gaussian field is optimized per sample by rendering it back to each camera view and matching the VGGT geometry and dense CLIP features:

$$(\hat{F}^v, \hat{D}^v, \hat{A}^v) = \mathcal{R}(\{g_i\}; K^v, E^v), \quad (3)$$

$$\mathcal{L}_{\text{fit}} = \lambda_f(1 - \cos(\hat{F}, F_{\text{vis}})) + \lambda_d\|\hat{D} - D\|_1 + \lambda_a\|\hat{A} - A\|_1 + \mathcal{L}_{\text{reg}}. \quad (4)$$

The fitted field is saved as a teacher cache. We store rendered targets rather than raw CLIP tokens:

$$T = \{T_{\text{dense3D}}, T_{\text{depth}}, T_{\alpha}, T_{\text{valid}}\}. \quad (5)$$

Here T_{dense3D} is the rendered Gaussian feature map. Although its semantic source is CLIP, we call it *dense3D* because the feature is lifted to a 3D Gaussian field and rendered with camera geometry. For multi-view policies, per-view targets are further packed into a fixed mosaic layout.

C. FastWAM-style Dual-Expert MoT

FastWAM-style models [8] use separate but interacting world/video and action experts. Gaussian4XWAM supervises the world/video branch while leaving the action prediction interface unchanged. Given world tokens h_{world}^ℓ at layer ℓ , a lightweight Gaussian head predicts the teacher targets:

$$(\hat{T}_{\text{dense3D}}, \hat{T}_{\text{depth}}, \hat{T}_{\alpha}) = \phi_G(h_{\text{world}}^\ell). \quad (6)$$

This encourages the world branch to encode 3D geometry, visibility, and dense semantic structure. Since the action branch consumes or interacts with the world representation, the enhanced representation can improve action learning without changing inference.

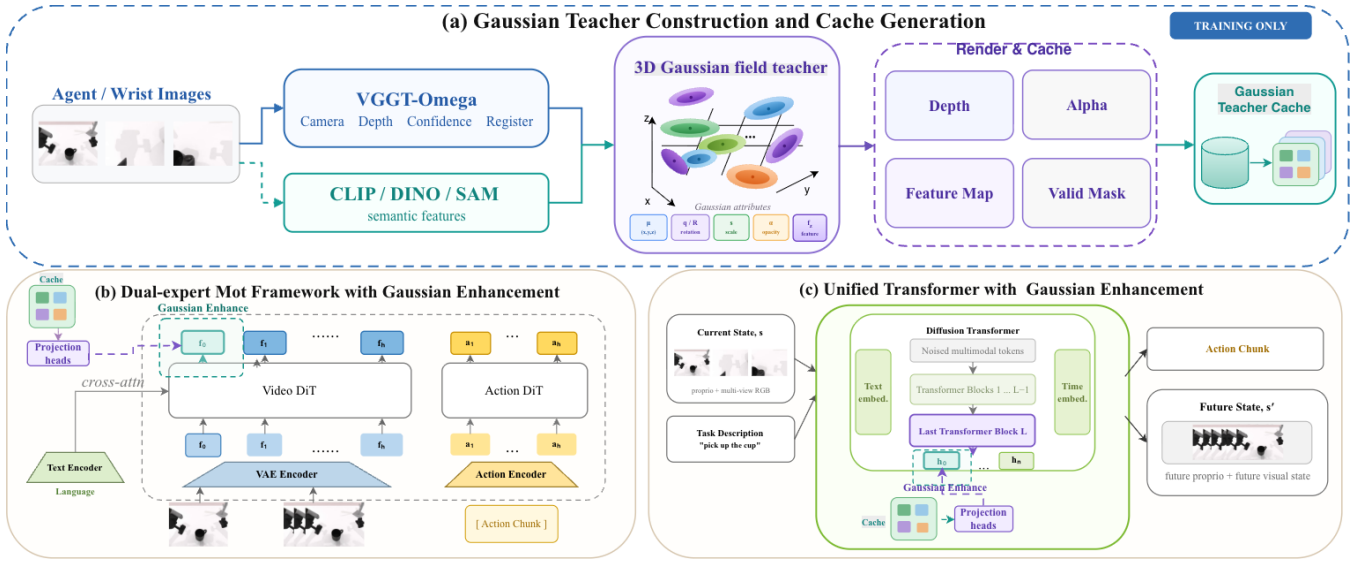


Fig. 2. Overview of Gaussian4XWAM. A 3D Gaussian teacher is constructed from multi-view robot observations and rendered into dense feature, depth, alpha/mask, and validity targets. These targets supervise action-relevant WAM representations during training. At inference time, the Gaussian teacher, renderer, and auxiliary heads are removed, leaving the original WAM policy unchanged.

D. Cosmos-Policy-style Unified Transformer

Cosmos-Policy-style models [9] use a unified Transformer for visual, world, and action-related tokens. We apply Gaussian4XWAM by supervising selected intermediate visual/world tokens. Let h^ℓ be the hidden states at layer ℓ . A Gaussian head predicts the same teacher targets:

$$(\hat{T}_{\text{dense3D}}, \hat{T}_{\text{depth}}, \hat{T}_\alpha) = \psi_G(h^\ell). \quad (7)$$

The supervised layer is configurable; in our main Cosmos experiments we use the final Transformer layer. Compared with the dual-expert setting, the Gaussian gradients act on a more shared backbone, but the framework remains unchanged: the teacher is only an auxiliary training signal and is removed at deployment.

E. Training Objective and Inference

Gaussian4XWAM augments the original WAM objective with masked Gaussian teacher losses. Let $\mathcal{L}_{\text{WAM}}$ be the base model loss. The teacher loss is

$$\mathcal{L}_G = \lambda_{\text{dense}} \mathcal{L}_{\text{dense3D}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_\alpha \mathcal{L}_\alpha, \quad (8)$$

where all terms are computed only on T_{valid} . The full training objective is

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{WAM}} + \mathcal{L}_G. \quad (9)$$

At inference time, the Gaussian cache, renderer, and teacher heads are discarded. The policy receives the same inputs as the original WAM and produces actions through the original forward pass, so Gaussian4XWAM adds no inference overhead.

III. EXPERIMENTS

A. Experimental Setup

We evaluate Gaussian4XWAM on LIBERO [10] and LIBERO-Plus [11]. These benchmarks cover language-conditioned manipulation under standard and distribution-shifted settings. We report success rate as the main metric.

a) LIBERO and LIBERO-Plus. LIBERO is used to evaluate language-conditioned manipulation under standard benchmark settings. LIBERO-Plus further stresses generalization by introducing harder task variants and distribution shifts, including camera, robot, language, lighting, background, noise, and layout variations. For both benchmarks, we compare FastWAM with the same backbone trained with Gaussian4XWAM teacher supervision.

b) Implementation. For each dataset, Gaussian teacher targets are precomputed offline and cached before policy training. During training, the teacher supervises selected internal world/visual tokens through dense3D, depth, alpha, and valid-mask targets. During evaluation, all teacher modules are removed, and the policy is executed with the same inference path as the original WAM.

B. Main Results

Table I summarizes the current results. Compared with the FastWAM baseline, Gaussian4XWAM improves the standard LIBERO success rate from 96.6% to 97.6%. On LIBERO-Plus, it improves the official overall success rate from 52.05% to 70.99%, with especially large gains under camera, robot, lighting, noise, and layout variations. We also observe consistent gains on the Cosmos Policy backbone under most robustness shifts after the same 5k-iteration training budget, improving the overall score from 71.52% to 77.30%.

TABLE I

MAIN RESULTS ON LIBERO AND LIBERO-PLUS. WE REPORT SUCCESS RATE (%). THE LIBERO-PLUS COLUMNS CORRESPOND TO SEVEN DISTRIBUTION SHIFTS. OVERALL IS COMPUTED OVER THE OFFICIAL 10,030 LIBERO-PLUS EVALUATION TASKS (CAMERA: 1599, ROBOT: 1550, LANGUAGE: 1537, LIGHT: 1142, BACKGROUND: 1076, NOISE: 1601, LAYOUT: 1525), RATHER THAN TAKING AN UNWEIGHTED MEAN ACROSS THE SEVEN CATEGORIES. EMB. PT. INDICATES WHETHER LARGE-SCALE EMBODIED PRETRAINING IS USED. OUR FASTWAM AND COSMOS POLICY BASELINES, AS WELL AS THEIR GAUSSIAN4XWAM VARIANTS, ARE TRAINED FOR 5K ITERATIONS.

Method	Type	Emb. PT.	LIBERO	Camera	Robot	Lang.	Light	BG	Noise	Layout	Overall
UniVLA [5]	VLA	Yes	95.20	1.80	46.20	69.60	69.00	90.70	21.20	31.90	43.90
π_0 [2]	VLA	Yes	94.20	13.80	6.00	58.80	85.00	90.70	79.00	68.90	54.60
π_0 -FAST [3]	VLA	Yes	85.50	65.10	21.60	61.00	73.20	97.70	74.40	68.80	64.20
OpenVLA-OFT [4]	VLA	Yes	97.10	56.40	31.90	79.50	88.70	97.30	75.80	74.20	70.00
X-VLA [6]	VLA	Yes	98.10	23.40	89.70	75.70	88.20	96.00	62.70	71.80	70.46
Spatial Forcing [7]	VLA	Yes	98.50	20.10	13.40	40.90	29.10	33.40	25.70	39.30	28.52
SG-WAM [17]	WAM	No	98.50	58.60	48.90	81.40	89.80	86.10	80.70	74.20	73.00
DreamWAM [18]	WAM	No	98.90	53.78	63.61	94.80	96.67	71.56	67.15	80.72	74.61
FastWAM [8]	WAM	No	96.60	25.63	42.88	73.88	61.70	54.88	43.55	66.80	52.05
FastWAM [8] + Gaussian4XWAM	WAM	No	97.60 (+1.00)	58.85 (+33.22)	63.74 (+20.86)	72.61 (-1.27)	87.74 (+26.04)	59.29 (+4.41)	79.51 (+35.96)	76.20 (+9.40)	70.99 (+18.94)
Cosmos Policy [9]	WAM	No	–	69.67	42.39	86.26	90.89	77.51	75.58	65.23	71.52
Cosmos Policy [9] + Gaussian4XWAM	WAM	No	–	79.11 (+9.44)	56.52 (+14.13)	92.18 (+5.92)	89.84 (-1.05)	66.17 (-11.34)	86.26 (+10.68)	70.55 (+5.32)	77.30 (+5.78)

TABLE II

TASK ABLATION STUDY ON LIBERO-PLUS WITH FASTWAM + GAUSSIAN4XWAM. WE REPORT SUCCESS RATE (%).

Variant	Camera	Robot	Lang.	Light	BG	Noise	Layout	Avg.
FastWAM	25.63	42.88	73.88	61.70	54.88	43.55	66.80	52.76
Full	58.85	63.74	72.61	87.74	59.29	79.51	76.20	71.13
No dense3D	49.30	62.80	73.30	84.80	55.40	77.20	74.60	68.10
No Depth	54.66	61.61	75.41	90.81	58.18	79.02	76.72	70.92
No Alpha	49.78	60.52	77.23	85.46	54.00	76.89	74.30	68.31

TABLE III

LAYER ABLATION STUDY ON LIBERO-PLUS WITH FASTWAM + GAUSSIAN4XWAM. LAYER 30 IS USED AS THE FINAL MODEL. WE REPORT SUCCESS RATE (%).

Layer	Camera	Robot	Lang.	Light	BG	Noise	Layout	Avg.
10	52.28	64.97	62.65	88.79	61.52	78.08	70.49	67.87
20	51.91	62.77	69.81	86.95	64.50	75.20	72.00	68.44
30	58.85	63.74	72.61	87.74	59.29	79.51	76.20	71.13

C. Ablation Study

a) *Discussion.*: The main results show that training-time Gaussian supervision is most beneficial under shifts that require explicit spatial reasoning. For FastWAM, Gaussian4XWAM yields large gains on camera (+33.22), robot (+20.86), lighting (+26.04), noise (+35.96), and layout (+9.40), improving the official LIBERO-Plus overall score by +18.94 points while also increasing the standard LIBERO score by +1.00. The Cosmos Policy results show the same trend under a different backbone: Gaussian4XWAM improves the overall robustness score by +5.78 points, with clear gains on camera, robot, language, noise, and layout. These improvements support the central hypothesis that a training-only 3D teacher can strengthen action-relevant world representations without adding inference-time cost.

The ablation results further clarify where the improvement comes from. Removing dense3D supervision or alpha supervision reduces the average performance from 71.13 to 68.10 and 68.31, respectively, indicating that both semantic 3D features and visibility-aware masks are important for robust manipulation. Removing depth still preserves a

strong average of 70.92, but changes the per-task behavior, suggesting that depth supervision is complementary rather than uniformly dominant. Finally, supervising deeper tokens works best overall: layer 30 achieves the highest average score, while earlier layers are better only for a small subset of shifts. This suggests that geometry supervision is most effective when applied to high-level action-relevant world tokens.

IV. CONCLUSIONS

We presented Gaussian4XWAM, a Gaussian field enhancement framework for world-action models. Instead of turning WAMs into explicit 3D or 4D world models, Gaussian4XWAM uses a 3D Gaussian field only as a training-time teacher. The teacher provides dense feature, depth, alpha/mask, and validity supervision to shape action-relevant latent representations, and is completely removed during inference. This design preserves the original WAM deployment path while injecting explicit 3D geometric structure during learning.

Experiments on LIBERO and LIBERO-Plus show that Gaussian4XWAM consistently improves both FastWAM and Cosmos Policy under the same 5k-iteration training budget. The gains are especially pronounced under camera, robot, noise, and layout shifts, where geometric grounding is critical. Ablations show that dense3D features, alpha supervision, and the choice of supervised layer all contribute to performance. These results suggest that training-only 3D supervision is a practical way to improve spatial grounding and robustness in WAMs without sacrificing inference efficiency.

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